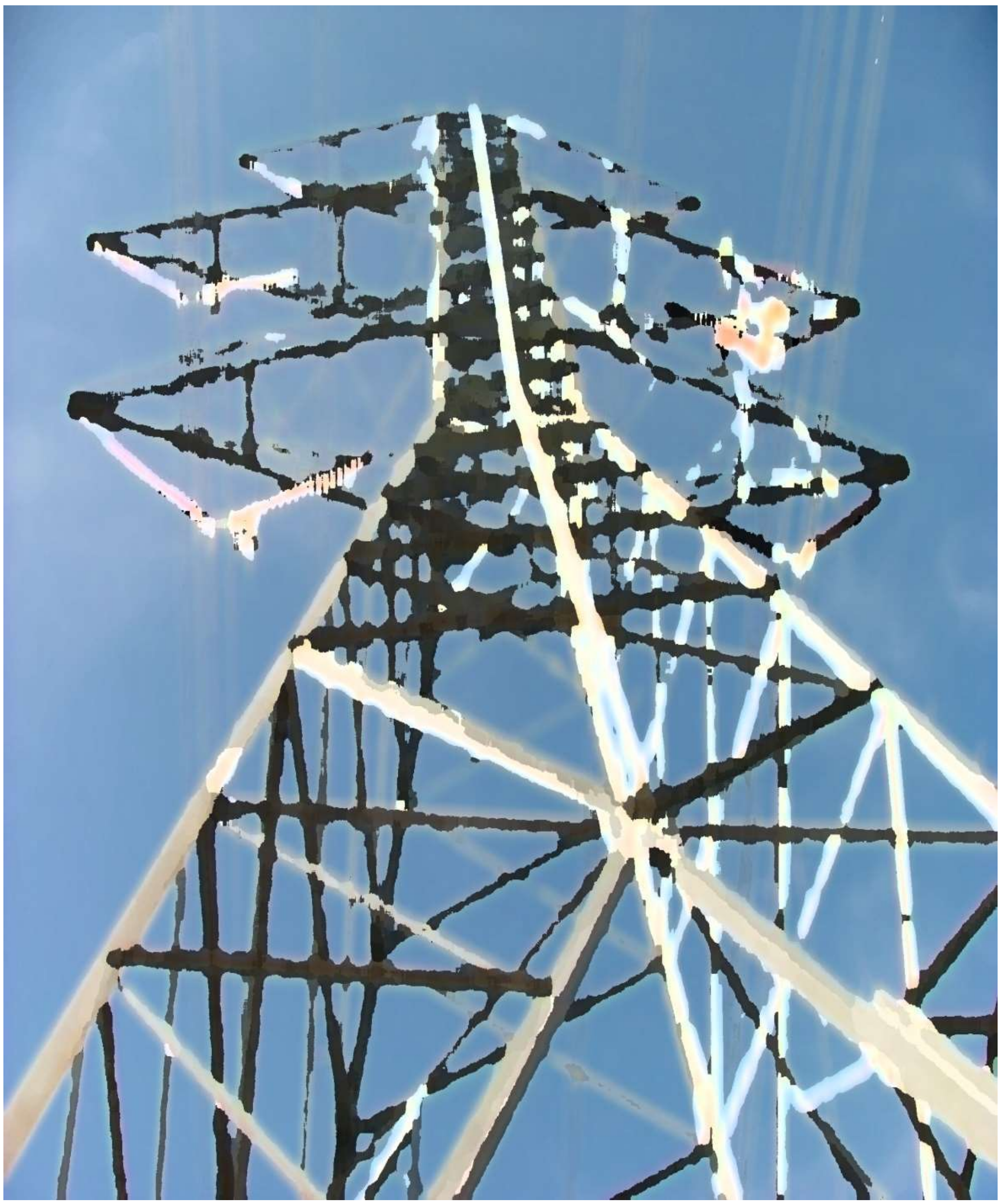
# Fall Protection

# Handbook



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## Table of Contents

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# SECTION I
## FALL PROTECTION ADMINISTRATION

## Dedication


DEDICATED TO
MATTHEW J. KARSTETTER
"Our Brother"
JANUARY 22, 1978
SEPTEMBER 20, 2012

## Fall Protection

Bonneville Power Administration (BPA) recognizes safety and health as a core value. BPA is committed to preventing fall-related incidents by following OSHA's best practices that utilize a fall protection hierarchy of fall hazard elimination or control. The following shall be considered in designing a fall protection solution: elimination or substitution, passive fall protection, restraint, fall arrest and administrative controls. First consideration shall be given to the elimination of fall hazards. Where elimination of the fall hazard is not practical, effective control of the fall hazard shall be used at all times.

Fall Protection at BPA is governed by the Fall Protection Committee. The mission of the Fall Protection Committee (FPC) is the continuous improvement of fall protection practices and policies to eliminate fall-related injuries and fatalities at BPA. The FPC has the delegated authority to set and change programs, procedures and processes for all matters related to fall protection. See charter for more specific information on the committee.

# SECTION II

## FALL PROTECTION INTERPRETATIONS, DEFINITIONS, POLICIES AND INFORMATION

### Interpretations

- “May”: Permissive choice (“may” equals “is permitted”).
- “Must” or “Shall”: Mandatory under normal conditions (“must” or “shall” equals “is required to”).
- “Should”: Advisory, “should” statements represent the best advice available at the time of printing (“should” equals “is recommended that”).

## Definitions

**Aerial Device:** Any piece of equipment using a bucket or platform to place the worker(s) at an elevated worksite.

**Anchorage:** A secure connecting point or a terminating component of a structure that is designed to support the potential load applied without failure.

**Anchorage Strength:** Strength of the structural member for fall arrest system **shall** be rated at 5000 Lbs. or two times the expected load for an engineered anchorage system. Work positioning devices **shall** be secured to (or around) an anchorage capable of supporting at least twice the potential impact load of a worker fall or 3,000 Lbs.

**Anchor Strap:** Sling or strap made from rope, webbing, wire or chain that is used to attach to a structure as an anchorage for an anchor line and that is rated for fall protection use.

**Authorized Person:** See SOHM Chapter 9.

**Back-up-device:** Also known as "**Fall Arrestor**", a device used on the safety line that tracks freely up or down as you ascend or descend the working line, automatically locking on the safety line when a sudden movement occurs.

**Carabiner:** A connector component generally comprised of a trapezoidal or oval shaped body with a normally closed gate or similar arrangement which may be opened to permit the body to receive an object and when released, automatically closes to retain the object.

**Climbing:** Vertical (ascending and descending) and horizontal movement to access the elevated worksite. (See "Transferring" and "Transitioning".)

**Competent Person:** See SOHM Chapter 9

**Deceleration Device**: A device which serves to dissipate the kinetic energy developed during free fall, or otherwise limits the energy imposed on an employee during fall arrest.

**Deceleration Distance**: The distance required for an energy absorbing device to dissipate the energy generated during a fall. This is accomplished through tearing, deformation or elongation of the energy absorbing device.

**Detachable Ladders:** Detachable ladders are those that are not permanently installed to a structure but are the normal means of accessing the work location on a structure or facility.

**Energy (shock) Absorber**: A component whose primary function is to dissipate energy and limit deceleration forces on the body during fall arrest. Such devices may employ various principles such as deformation, friction, tearing of materials or breaking of stitches to accomplish energy absorption. An energy absorber causes an increase in the deceleration distance. Shock-absorbing lanyards **shall** only be used for fall arrest. They are not designed for and **shall not** be used for work positioning.

**Engineered Anchorage:** A fall protection anchorage which is designed and will operate to withstand the maximum expected impact load while maintaining a specified overload capacity factor (OCF) of two. (Example: if a maximum arresting force (MAF) of 1200 Lbs. is expected with a particular fall arrest system, the engineered anchorage must be certified to withstand at least 2,400 Lbs. of load.)

**Engineered System:** A fall protection system which is designed with the intent of protecting an employee at height.

**Factor of 2 Fall:** A fall where the distance traveled before the fall protection system engages is twice the length of the fall protection system. (i.e.) when the anchorage is located at your feet.

**Fall Arrestor:** A deceleration device which travels on a lifeline and automatically engages the lifeline and locks to arrest the fall of an employee.

**Fall Arrest System:** A system designed to protect the worker by limiting the arresting forces generated during a fall.

**Fall Prevention System:** A system intended to prevent a worker from falling from one elevation to another. Such systems include guardrails, barriers, parapets, covers, and hatches. Fall prevention systems act as a physical barrier preventing a worker from reaching a fall hazard.

**Fall Protection:** Any equipment, device or system that prevents a fall from elevation or that mitigates the effect of such a fall.

**Fall Protection Plan:** A plan that shall be developed and incorporated into the J1 or JHA that identifies fall hazards and mitigations.

**Fall Protection Program:** A comprehensive managed fall protection program is the most effective method to: identify, evaluate, and eliminate (or control) fall hazards through planning; ensure proper training of workers exposed to fall hazards; ensure proper installation and use of fall protection and rescue systems; and implement safe fall protection and rescue procedures.

**Fall Restraint System:** A physical connection between an anchor and the worker using equipment that will allow the worker to approach a hazard but prevents them from falling into the hazard.

**Free Fall Distance:** The vertical distance traveled during a fall, measured from the onset of the fall to the point the fall arrest system begins to arrest the fall.

**Full-Body Rescue Harness:** A device that is designed to encapsulate a worker and transfer the forces generated during a fall throughout the body. The harness can be equipped with a connection point that will allow for the connection of fall protection components. That will facilitate safe work practices and rescue.

> **NOTE:** Wherever the word "harness" is used by itself in this handbook, it refers to full-body rescue harness unless otherwise specified.

**Hazard:** An event which can potentially endanger personnel, impair safe working conditions, and has the potential to cause injury or loss of life.

**Horizontal Lifeline:** A system designed by a qualified person consisting of components configured between two anchorages that allows horizontal movement between two points.

**J1 Job Briefing:** See APM, J1 rule.

**Jobsite:** The assembly point at the structure or equipment where the workers, tools and vehicles are assembled to perform the climbing to the worksite.

**Kilo Newton (kN):** Metric unit of force where 1 kN = 224.8 Lbs.

**Ladder: (Fixed or Portable)** A structure consisting of a series of bars or steps between two upright lengths of wood, metal or rope. Used for climbing up or down something.

**Lanyard:** A flexible line of rope or strap which generally has a connector at each end for connecting to a full-body rescue harness and to a deceleration device, lifeline, or anchorage.

**Layback:** The portion of a tower that lays outward from the waist creating a window for the center phase. Step bolts will be in line on only one side of the angled steel member.

**Legacy Structures**: Structures that were in service on BPA's system prior to recognizing the need to design fall protection into the structures.

**Maximum Arresting Force:** The Maximum Arrest Force (MAF) to which a user of a Fall Arrest System that includes a full body harness is exposed to during the arrest of the fall. The MAF **shall** be limited to 1,800 Lbs. in fall arrest and 900 Lbs. in work positioning. (Double skidding would potentially exceed this requirement.)

**Overload Capacity Factor (Safety Factor):** The number by which a maximum load is multiplied to assure that the system does not fail when loaded to the design load.

**Pelican Hook:** A snap hook style connector with a large gate opening.

**Pole Strap:** See "Positioning Strap".

**Positioning Strap:** A strap with locking snap hook(s) or carabiner to connect to the side D-rings of a worker's full-body rescue harness. Used as a positioning device (also known as pole strap, scare strap or safety strap).

**Qualified Person:** See SOHM Chapter 9.

**Roll-Out:** A movement or process by which a snap hook or carabiner unintentionally disengages from an anchorage or object to which it is coupled.

**Rope Access:** Rope access is a form of vertical access, initially developed from techniques used in climbing and caving, which applies practical rope work to allow workers to access difficult-to-reach locations without the use of scaffolding, cradles or an aerial work platform. Rope access technicians descend, ascend, and traverse ropes for access and work while suspended by their harness. This system incorporates two independent safety ropes.

**Rope Grab:** A device that holds position on a rope until intentionally disengaged, generally seen where the connector(s) are unable to maneuver due to a rigged connection.

**Safety Strap:** See “Positioning Strap”.

**Scissor Lift:** A type of platform that can usually only move vertically. The mechanism to achieve this is the use of linked, folding supports in a crisscross "X" pattern, known as a pantograph (or scissor mechanism).

**Secondary Lanyard:** An adjustable rope or strap used as a work positioning device and to pass obstacles to maintain 100% attachment. While using a secondary lanyard, climbers must stay within the 2-foot fall distance allowed while in work positioning.

**Secondary Fall Protection System:** One or more means of fall protection, configured as a supplement or as backup to protect a worker from a potential fall if the primary system fails.

**Self-Retracting Device (SRD or SRL):** A device which contains a drum-wound web lanyard or steel line which may be slowly extracted from or retracted onto the drum under slight tension during normal movement of the user. The line has means for attachment to the fall arrest attachment on the body support. After onset of a fall, the device automatically locks the drum and arrests the fall. The device may have integral means for energy absorption.

**Snap Hook**: A connector comprised of a hook-shaped member with a normally closed keeper or similar arrangement, which may be opened to permit the hook to receive an object and when released, automatically closes to retain the object.

**Suspension Trauma**: (orthostatic shock while suspended), is an effect which occurs when the human body is held upright without any movement for a period of time. If the person is strapped into a harness or tied to an upright object, they will eventually suffer the central ischemic response (commonly known as fainting).

**Suspension Trauma Straps:** Rescue straps that are attached to a harness and deployed by a worker after a fall. The straps allow the worker to step into them and use their legs to support body weight.  Using the leg muscles helps to return venous blood in the legs back to the torso and promotes regular blood flow through the body.

**Swing Fall:** A pendulum like motion that occurs during and/or after a vertical fall. A swing fall results when a worker begins a fall from a position that is located horizontally away from a fixed anchorage.

**Total Required Distance:** The total distance required for the Fall Arrest system to protect a worker in the event of a fall.

**Transferring:** The act of moving from one distinct object to another (e.g., between an aerial device and a structure).

**Transitioning:** The act of moving from one location to another on equipment or a structure while going around or over an obstruction.

**Vertical Lifeline (VLL):** A component, element or constituent of a lifeline subsystem consisting of a vertically suspended flexible line and along which a fall arrester travels.

**Warning Line System:** A barrier erected on a roof to warn workers that they are approaching an unprotected roof side or edge, and which designates an area in which roofing work may take place without the use of guardrail, or safety net systems to protect workers in the area.

**Wood Pole Fall Restricting Device (WPFRD):** A wood pole climbing device designed to limit a fall to two feet or less on wood poles. WPFRD shall be used for climbing all wood structures at BPA.

**Work Positioning:** A system a worker uses to stay connected to a structure while freeing their hands to work. Work positioning consists of a full-body rescue harness with work positioning D-Rings and a lanyard around an anchor or structure while connecting to both side D-rings.  Do not connect your position strap, or rope lanyard around the anchor in a choker style or connect back into the same D-ring.

# Policy

## Fall Protection

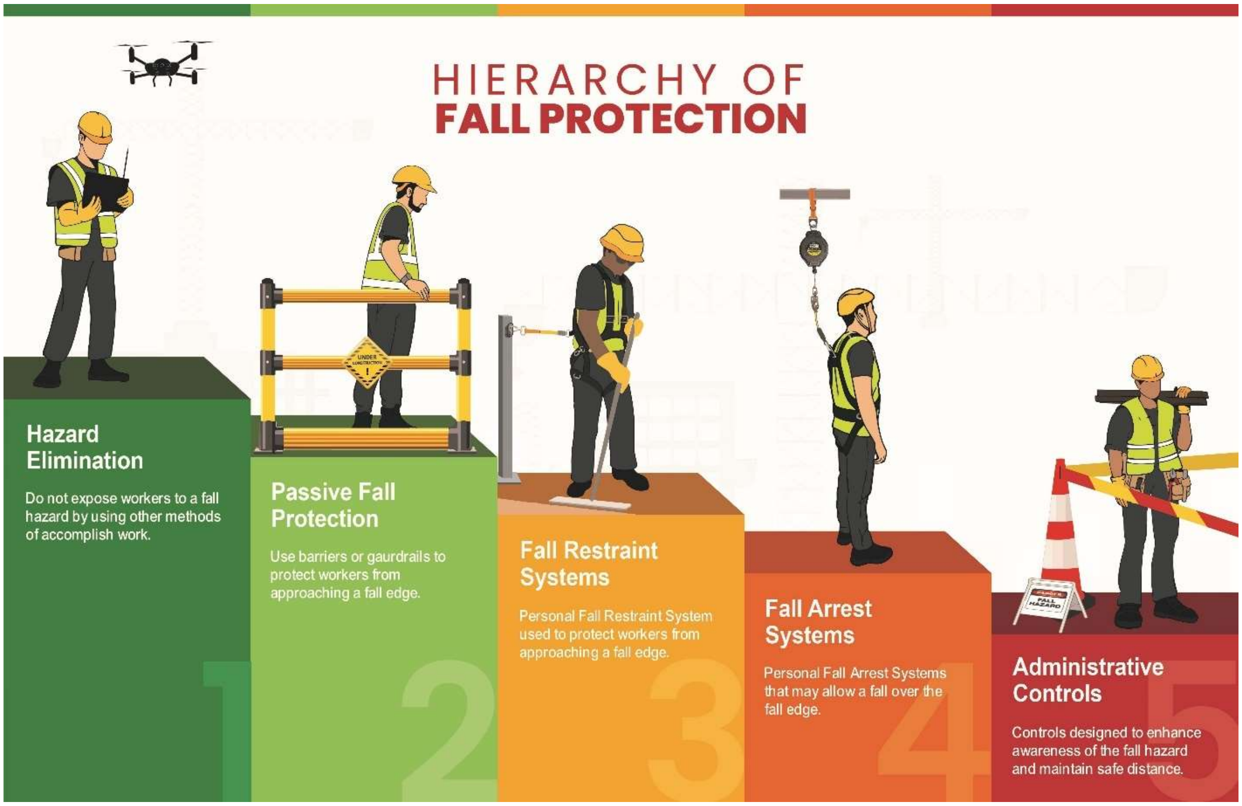
Fall protection **shall** be used by anyone working at elevated locations 4 feet or more above a lower level or hazard. Fall protection is not limited to just equipment that is worn by the worker. The term fall protection refers to any method of mitigating falls. A complete fall protection system will consist of both a primary and a secondary system. Where the primary system will provide support to the worker, the secondary system will provide a backup if the primary system fails.

During work activities above four feet, at least two trained workers **shall** be present and a means of rescue readily accessible. Workers who are on light duty or cannot physically perform the climbing activities for the day **shall** not be counted as one of the two trained workers.

## Fall Protection Plan

Prior to any work being performed, J1 and/or JHA **must** be conducted to identify the method of protection that would be advantageous to perform the work. Every effort **must** be taken to eliminate the fall hazards. Additionally, rescue and evacuation **must** be evaluated for every job where work is to be performed aloft or when work is performed where the only means of access or egress is vertical. Once the method of protection and rescue/egress has been determined, all workers involved **must** be aware of hazards associated with the work and trained.

## Fall Protection Hierarchy




When work is to be performed at heights that will require fall protection, an evaluation **shall** be conducted to determine the optimal means of protecting the employee(s) from the hazards. Industry best practices have recognized a hierarchy of controls to determine the best form of protection for an identified hazard.

### Hazard Elimination

A means of preventing an employee from being exposed to a hazard, this is generally done by engineering alternative means of conducting the work that eliminates exposure or eliminating the need to perform the work in the hazardous area by removing the hazard.

**Passive Fall Protection:** These systems create a physical barrier between the hazard and the employee. Systems like guardrails, parapets, safety nets and ladder cages are examples of passive fall protection. The systems need very little training to be effective.

**Fall Restraint Systems:** Used to protect workers from approaching a fall edge.

**Fall Arrest Systems:** These systems will allow a fall but are designed to arrest the fall without injury due to the force of the fall.

## Administrative Controls

A means of reducing an employee's exposure to a hazard, this can be accomplished by providing training that provides tools to employees to recognize potential hazards in a work area and create methods of controlling potential hazards. Administrative controls may be used to control access to work areas and set requirements for access of locations.

## Rescue

A means of rescue for a worker that is working aloft **shall** be readily available at the work location. A worker trained in rescue aloft shall be onsite and available to provide rescue during work aloft activities. In the event only one worker is capable to providing rescue, that worker shall not work aloft. In the event of a fall, the following methods of rescue shall be considered:

- **Self-rescue** – Worker egresses or climbs to a safe location under own power.
- **Assisted rescue** – When it has been determined that a worker is incapable of self-rescue, a rescuer shall assist or provide rescue to a worker. Rescuer shall evaluate the situation and determine if rescue will result in undue risk to self-prior to rescue attempt. A rescuer shall use back-up (safety system) while conducting rescue.
- **Suspended rescue** – In the event that a rescuer is unable or greater risk exists, a rescuer may use suspended rescue techniques to rescue or evacuate.

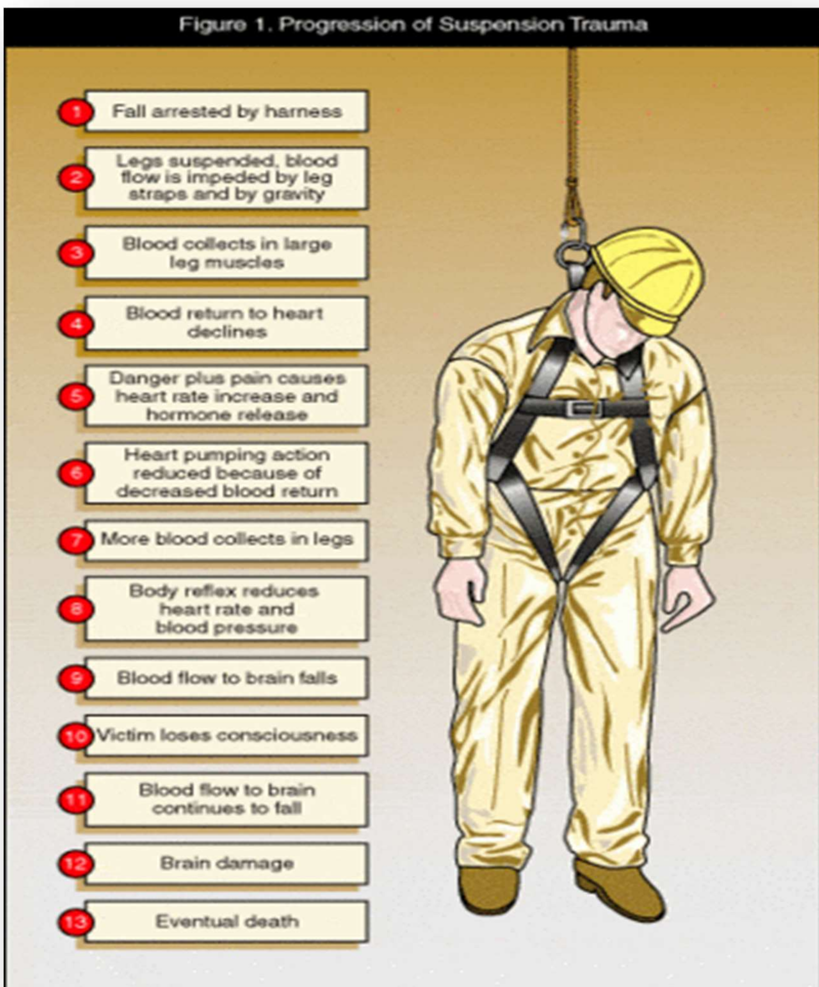
## Suspension Trauma

OSHA recommends the following general practices/considerations:

- Rescue suspended workers as quickly as possible
- Be aware that suspended workers are at risk of suspension trauma
- Be aware of signs and symptoms of suspension trauma
- Be aware that suspension trauma is potentially life threatening. Suspended workers with head injuries or who are unconscious are particularly at risk

## Signs of suspension trauma

- Faintness
- Breathlessness
- Sweating
- Paleness
- Hot Flashes
- Change in Heart Rate or blood pressure
- Nausea
- Dizziness
- “Graying” or Loss of Vision



## Suspension Trauma Straps

Suspension trauma straps **shall** be included on all harnesses. The trauma straps will be fixed to the harness on a load bearing point of the harness. The trauma straps will typically only work on a conscious fall victim. These straps allow the fall victim to alleviate symptoms of suspension trauma.

## Storm Damage

Thorough inspection of suspected or visually damaged facilities **shall** precede any climbing or work activity. Stresses that may have exceeded design limitations require engineering evaluation. Climbers **shall** inspect all climbing surfaces for damage to permanently attached facilities such as step bolts, rungs, attachment fixtures, and welds. In particular, the structural integrity of the anchorages for attaching work positioning and fall protection systems **shall** be scrutinized. In addition, other hazards (trees, buildings, adjacent structures, etc.) that may fall into or affect the structure's stability **shall** be analyzed before work is started on the structure.

## Inclement Weather

Weather conditions may adversely affect climbers and climbing surfaces. Prevailing local conditions (i.e., wind, ice, rain, and slippery surfaces) **shall** be considered in the selection of climbing methods and equipment. Actions **shall** be taken to minimize the effects of wet, icy, muddy, or slippery handholds and stepping surfaces (i.e., ice can be chipped or wait for conditions to improve). Towers are designed with a safety factor of 1.0 under extreme conditions. When a structure has ice or heavy snow loading, engineering evaluation is strongly recommended to verify design loading criteria. The additional loading of a fall arrest scenario or even just the additional weight of a worker may cause a structural overload condition.

## Equipment Inspection

Trained workers **shall** be competent as to the inspection criteria of the equipment provided and will inspect all fall protection equipment prior to each use. Fall protection and climbing equipment **shall** be stored and protected when not in use and **shall** be used in accordance with the manufacturer's recommendations and instructions. A competent detailed inspection of all fall protection equipment and systems will be conducted annually.

### Removal From Service: Wear, Damage or Age (based on manufacture guidance)

Fall protection equipment **shall** be removed from service when by inspection it is deemed to be deficient, damaged or compromised in any way that would affect its designed use. Once removed from service the piece of equipment **shall** be destroyed to prevent accidental future use. (With the exception of rescue systems and SRD's, this may be serviced by the manufacturer.)

### Removal From Service if Involved in an Actual Fall

Should a fall protection system (including the anchor) be involved in an actual fall arrest scenario, where deployment of impact indicators or energy absorber has taken place. The components of the system **shall** be removed from service and sent to the fall protection program manager; the anchor shall be inspected by a qualified person.

### Rescue Equipment Maintenance Cycle

Rescue equipment **shall** be inspected annually by a competent person and serviced in accordance with the manufacturer guidelines. It is recommended that the rescue equipment is stored in such a way that a tamper seal can be placed on the kit ensuring all necessary components remain in the kit.

### Training

A worker whose job title requires work to be done at elevated work locations above four feet, **shall** be trained in fall protection practices and rescue procedures to a level that is equivalent to the highest exposure they will face. The training will be developed based upon practices within this policy and from industry best practices. Training curriculum shall be evaluated annually to ensure the needs of stakeholders are being met and to stay up to date with technology and techniques.

#### Fall Protection Training

Training shall provide both ideological and tangible understanding of this policy and the tools and equipment available to workers. The training will be balanced between classroom and hands-on time. The training will focus on teaching workers to:

- Identify Hazards and plan a corrective action
- Interpret BPA Fall Protection Policy
- Utilize equipment available to workers
- Create site specific rescue plans
- Provide rescue and egress when necessary
- Competently inspect equipment

Upon completion of the course, workers will be able to conduct work at heights within the scope of this policy and **shall** retrain annually to refresh proficiencies with equipment and policy.

The following courses address Authorized Users, Competent Persons, and Rescue training requirements based on the fall exposures of these employees.

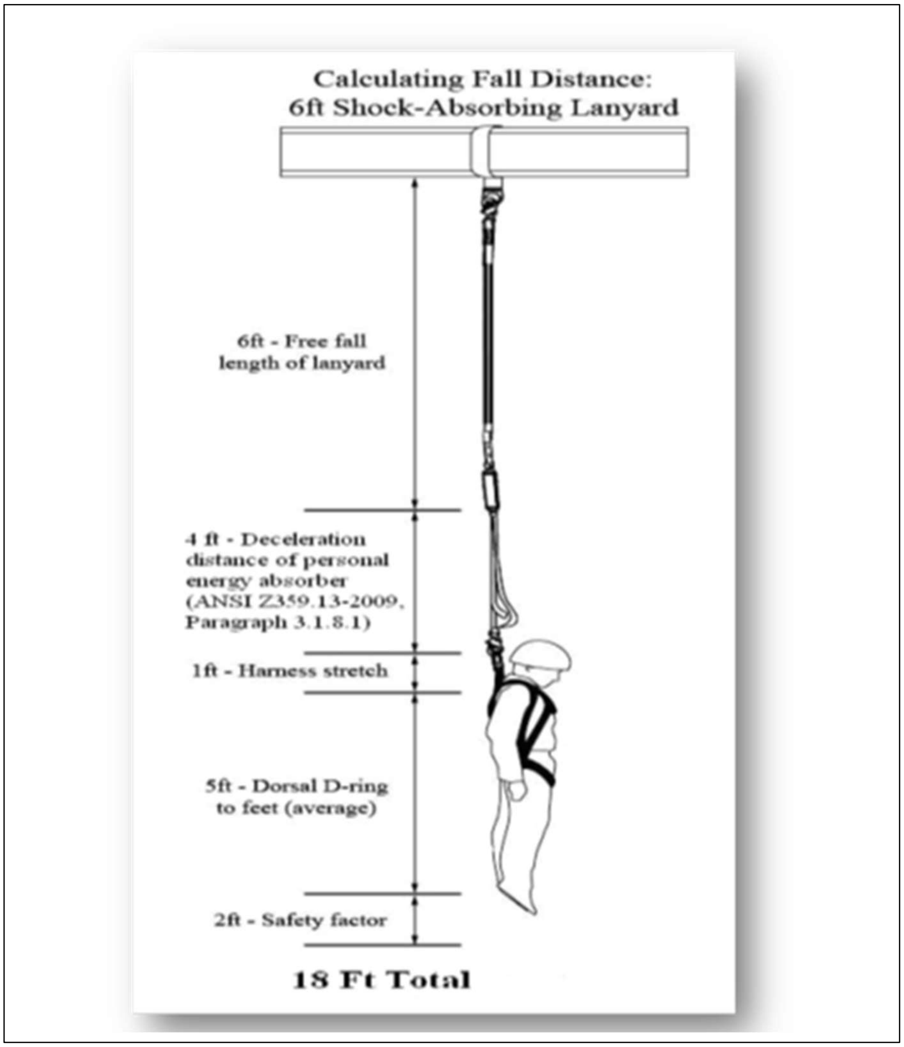
| | | | | |
|---|---|---|---|---|
| BPA-3013-TG-FPAU | Fall Protection Authorized User | 8 | Harness don, doff, and fit<br>Rope Grab use and limitations<br>SRL use and limitations (only if used)<br>Work Positioning Lanyard (only if used)<br>Bucket Truck Self Rescue (only if used) | Operators<br>EP<br>Warehouse<br>HMEM |
| BPA-3014-TG-FPCP-ELC | Fall Protection Competent Person | 12 | Equipment Inspection<br>Ropes and Knots<br>Fall Arrest<br>Fall Restraint<br>SRLs<br>Positioning Systems<br>Bucket Truck Bailout<br>Hanging Victim Rescue<br>Tracking Line Rescue | New Hires<br>Electricians<br>Crafts People<br>Painters<br>Welders |
| BPA-3015-TG-FPCP-R-ELC | Fall Protection Competent Person Refresher | 8 | Equipment Inspection<br>Ropes and Knots<br>Fall Arrest<br>Fall Restraint<br>SRLs<br>Positioning Systems<br>Bucket Truck Bailout<br>Hanging Victim Rescue<br>Tracking Line Rescue | Electricians<br>Painters<br>Welders |
| BPA-3016-TG-FPCP-TLM | Fall Protection Competent Person - TLM | 16 | Equipment Inspection<br>Ropes and Knots<br>Fall Arrest<br>Fall Restraint<br>SRLs<br>Positioning Systems<br>Rope Access Equipment<br>Bucket Truck Bailout<br>Pole Top Rescue<br>Tower Rescue | Lineman New Hires and Lineman Apprentices |

| | | | | |
|---|---|---|---|---|
| BPA-3017-TG-FPCP-R-TLM | Fall Protection Competent Person Refresher - TLM | 8 | Equipment Inspection<br>Ropes and Knots<br>Fall Arrest<br>Fall Restraint<br>SRLs<br>Positioning Systems<br>Rope Access Equipment<br>Bucket Truck Bailout<br>Pole Top Rescue<br>Tower Rescue | Lineman and Lineman Apprentices |

## Calculating Clearances

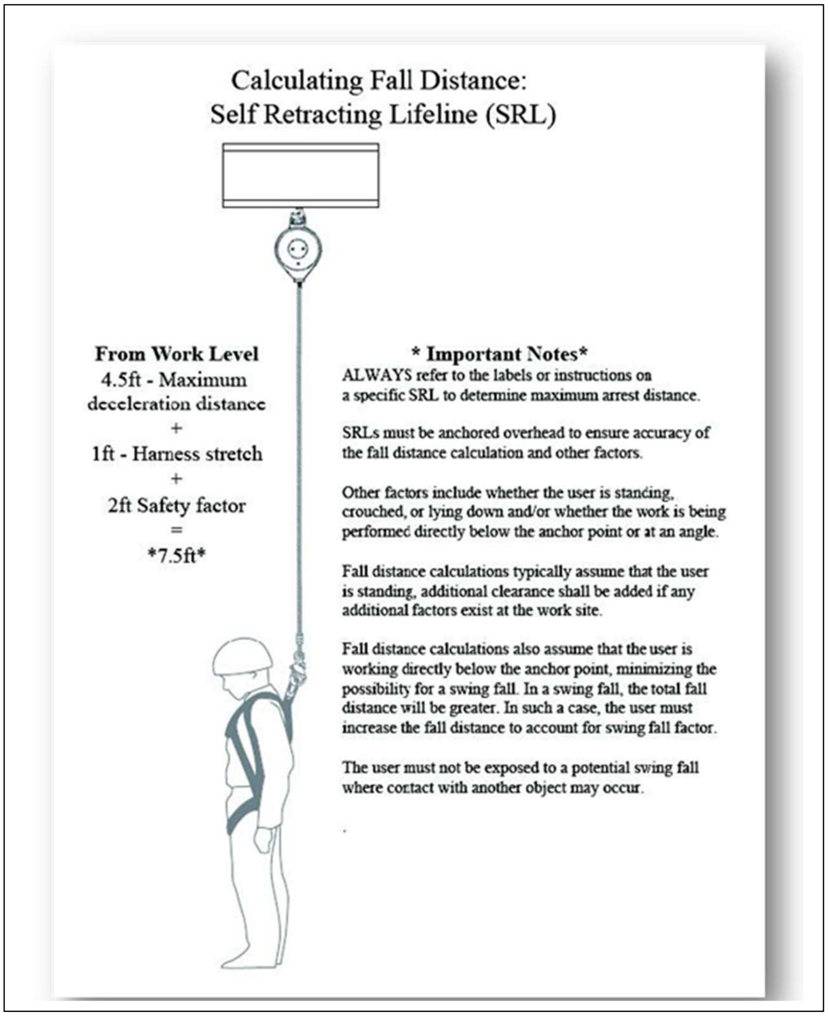
When using fall arrest systems, it is important to take into consideration the amount of clearance available to the nearest hazard in the fall path. A general rule to follow for fall arrest systems is as followed:

## Energy Absorbing Lanyard




The industry standard 6' free-fall energy absorbing lanyard will have a required clearance of 18' from the anchor point. This required clearance can be reduced by limiting the length of the lanyard or by reducing the free fall distance.

## Self-Retracting Devices

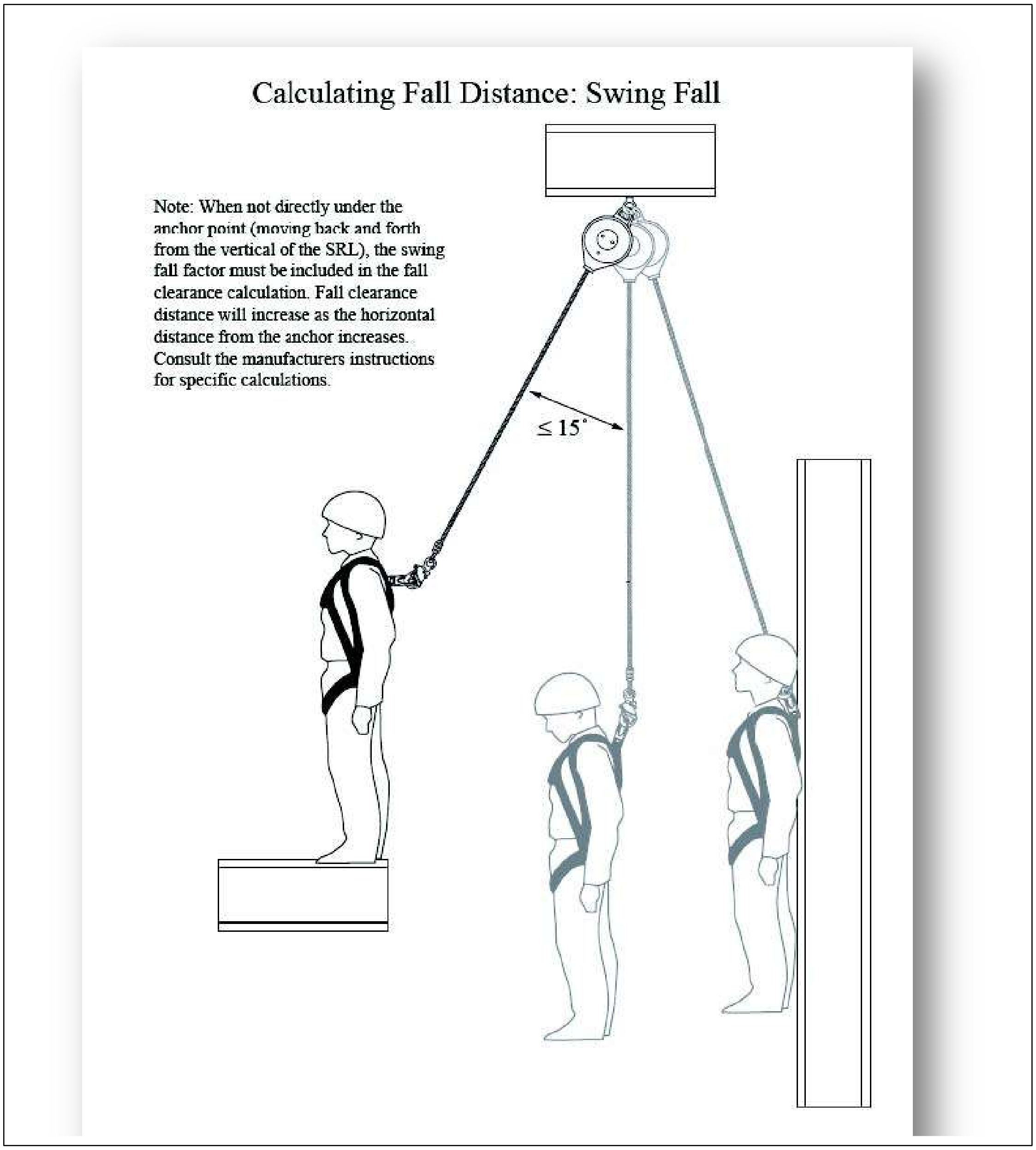



- Class 1 SRDs should only be used where the anchorage is located at a sufficient height above the walking/working surface such that the free fall distance should be two feet or less, and free fall and swing fall should not result in contact with a lower level or an object in the fall or swing path.

- Class 2 SRDs may be used in applications where free fall may exceed six feet and contact with the structural edge is possible. These conditions should be eliminated, if feasible due to the vulnerability of the line material to cutting and abrasion in conditions where sharp or abrasive edges may be present.

**NOTE:** Be aware of the swing fall that can occur during a fall and use the system within the manufactures specifications.

**Swing Fall** (See ANSI Z359.6-2009 for detailed procedures to calculate fall distance).



## System Selection Criteria

### Passive Fall Protection

Guardrails **shall** be 42” in height and capable of withstanding 200lbs of force in any direction. The mid-rail should be half the height of the top rail or 21" and have a 3.5” toe board to prevent tools and equipment from falling to the lower level.

All other covers **shall** be capable of supporting, without failure, at least twice the weight of workers, equipment, and materials that may be imposed on the cover at any one time. All covers shall be color coded, or they shall be marked with the word "HOLE" or "COVER" to provide warning of the hazard.

## Fall Restraint

Shall be rigged in such a way that the system will allow the worker to approach a hazard or edge but will not allow them to enter the hazard or pass the edge. Passive fall protection systems **shall** have manual rope grabs or adjusters that must be intentionally disengaged to be adjusted and **shall** have anchor strength of 1000lbs or a 2:1 safety factor of the possible intended load; whichever is greater **shall** be used. Care should be taken while working in proximity to corners where the system may not prevent a fall from occurring.

## Fall Arrest

Personal fall arrest systems and their use shall comply with the provisions set forth below. Effective January 1, 1998, body belts are not acceptable as part of a personal fall arrest system. Anchorages used for attachment of personal fall arrest equipment shall be independent of any anchorage being used to support or suspend platforms and capable of supporting at least 5,000 pounds (22.2 kN) per worker attached. Personal fall arrest systems shall limit the force transferred to a worker to 1800lbs and limit free fall to 6ft. Special attention must be taken to ensure adequate distance is available to allow proper deployment of the fall arrest system.

## Administrative Controls

1910.28(b)(13)(iii)
When work is performed 15 feet (4.6 m) or more from the roof edge, the employer must:
1910.28(b)(13)(iii)(A)
Protect each employee from falling by a guardrail system, safety net system, travel restraint system, or personal fall arrest system or a designated area. The employer is not required to provide any fall protection, provided the work is both infrequent and temporary; and
1910.28(b)(13)(iii)(B)
Implement and enforce a work rule prohibiting employees from going within 15 feet (4.6 m) of the roof edge without using fall protection in accordance with paragraphs (b)(13)(i) and (ii) of this section.

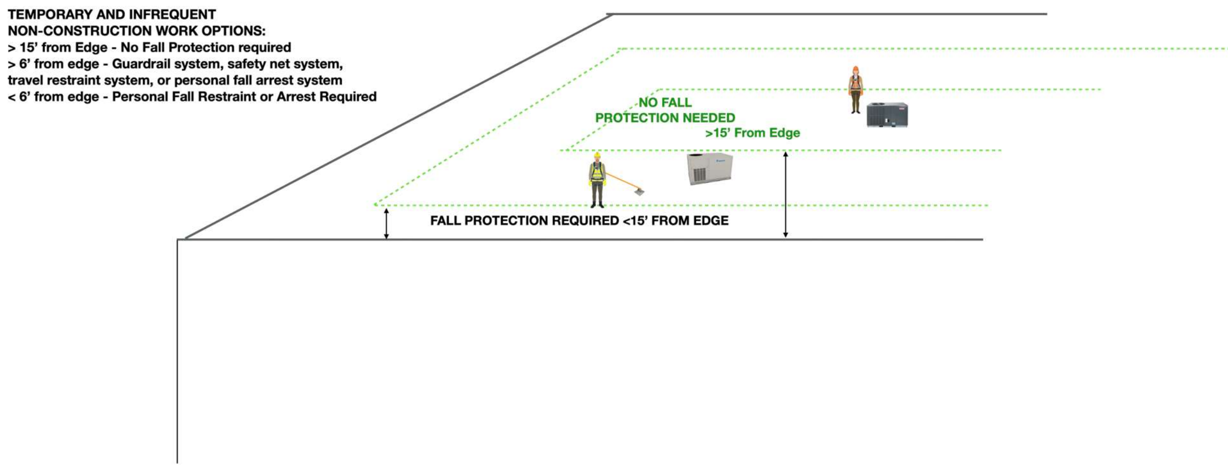
See the letter of interpretation.
https://www.osha.gov/laws-regs/standardinterpretations/2005-01-03

The warning line shall be erected around all sides of the roof work area.

For roofing work, when mechanical equipment is not being used, the warning line shall be erected not less than 6 feet (1.8 m) from the roof edge. When mechanical equipment is being used, the warning line shall be erected not less than 6 feet (1.8 m) from the roof edge which is parallel to the direction of mechanical equipment operation, and not less than 10 feet (3.1 m) from the roof edge which is perpendicular to the direction of mechanical equipment operation. Points of access, materials handling areas, storage areas, and hoisting areas shall be connected to the work area by an access path formed by two warning lines. When the path to a point of access is not in use, a rope, wire, chain, or other barricade, equivalent in strength and height to the warning line, shall be placed across the path at the point where the path intersects the warning line erected around the work area, or the path shall be offset such that a person cannot walk directly into the work area.

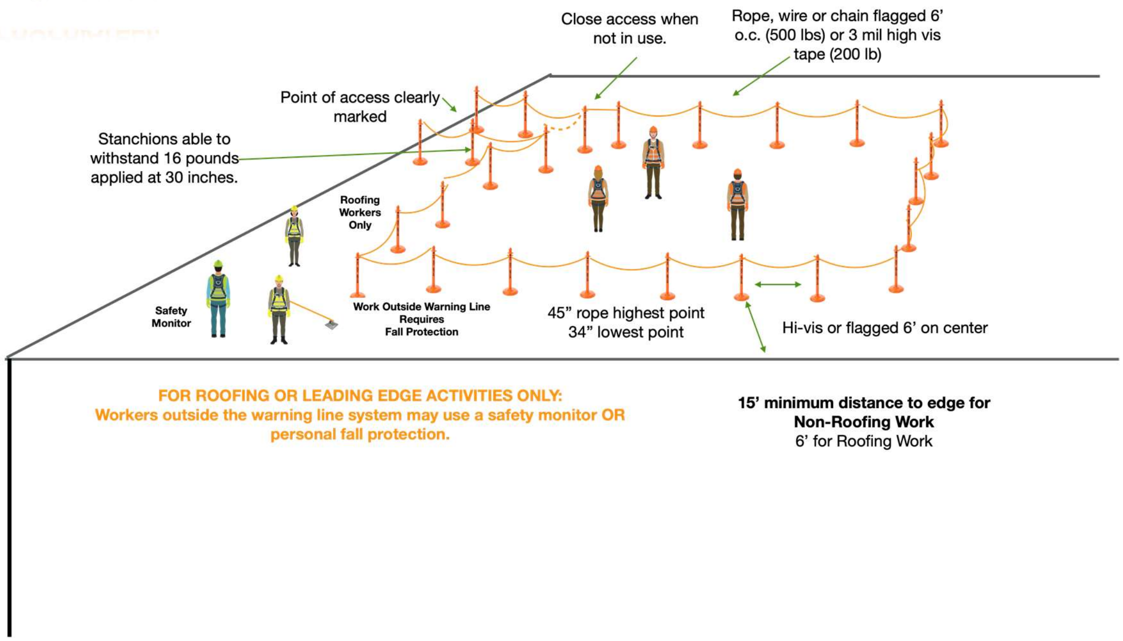
Warning lines shall consist of ropes, wires, or chains, and supporting stanchions erected as follows: the rope, wire, or chain shall be flagged at not more than 6-foot intervals with high-visibility material; The rope, wire, or chain shall be rigged and supported in such a way that its lowest point (including sag) is no less than 34 inches from the walking/working surface and its highest point is no more than 39 inches from the walking/working surface; After being erected, with the rope, wire, or chain attached, stanchions shall be capable of resisting, without tipping over, a force of at least 16 pounds applied horizontally against the stanchion, 30 inches above the walking/working surface, perpendicular to the warning line, and in the direction of the floor, roof, or platform edge; The rope, wire, or chain shall have a minimum tensile strength of 500 pounds and after being attached to the stanchions, shall be capable of supporting, without breaking, the loads applied to the stanchions as prescribed in paragraph of this section; and The line shall be attached at each stanchion in such a way that pulling on one section of the line between stanchions will not result in slack being taken up in adjacent sections before the stanchion tips over. No employee shall be allowed in the area between a roof edge and a warning line unless the worker is performing roofing work in that area.

## TEMPORARY AND INFREQUENT WORK OPTIONS




## WARNING LINE SYSTEMS

**VERY SAFE IF DESIGNED AND USED APPROPRIATELY.**



## Safety Monitor

Safety monitors are another way of warning workers of fall dangers.

The employer shall designate a competent person to monitor the safety of other workers and the employer shall ensure that the safety monitor complies with the following requirements: the safety monitor shall be competent to recognize fall hazards; warn workers when it appears that they are unaware of a fall hazard or is acting in an unsafe manner; be on the same walking/working surface and within visual sighting distance of the worker being monitored; be close enough to communicate orally with the worker; and not have other responsibilities which could take the monitor's attention from the monitoring function.

## Equipment

The equipment and products listed in the following section are examples and are subject to change due to technological advances or changes in technique. BPA is constantly evaluating equipment and techniques to provide the best equipment possible to workers. **Work performed within the scope of 1910.269 and this handbook requires the use of Arc Flash rated PPE.** (Note that all equipment shall be used within the scope of the manufactures guidelines.)

## Full Body Harness

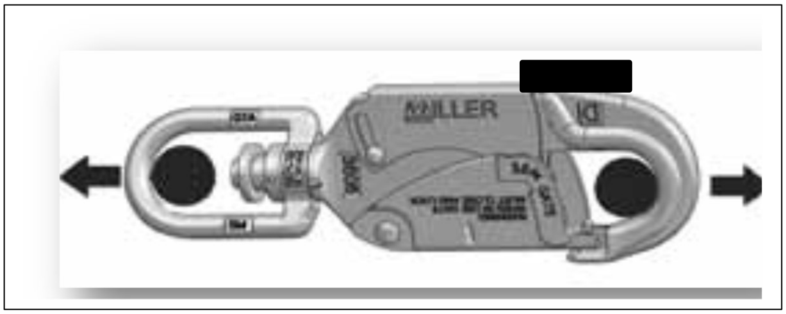
The full body harness provides a worker with a means of connecting into a fall protection system. There are many types of harnesses available, and the harness worn should be chosen based upon the work to be performed. In most cases a fall arrest harness like the one pictured above will provide both connection points and comfort. The fall arrest harness works by distributing the forces generated during a fall through the body where it is caught by the sub-pelvic strap. These harnesses keep the body upright during a fall which reduces the chances of secondary injuries and promotes a prompt rescue. All full body harnesses shall meet OSHA and ANSI Z359.11 requirements.

## Carabiners

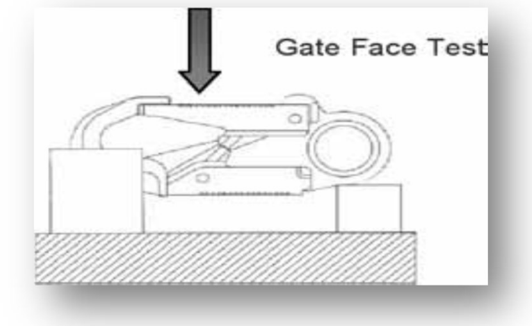
All carabiners used for fall protection/rescue **shall** be of the locking type with a self-closing, self-locking gate which remains closed and locked until intentionally unlocked and opened for connection or disconnection. Carabiners shall meet OSHA and ANSI Z359.12(09) requirements.

## Snap Hooks

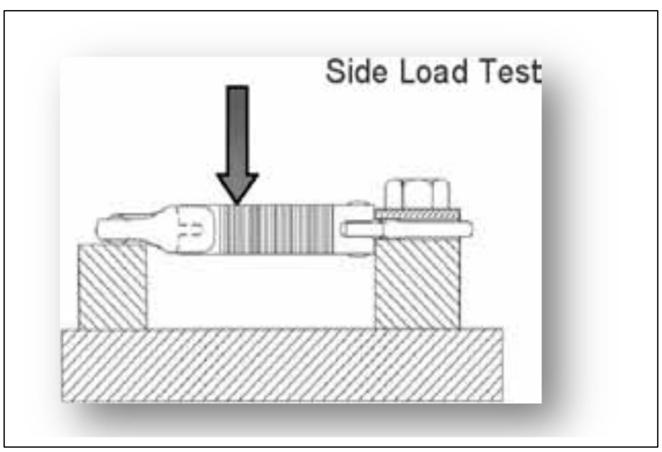
5,000 Lb. (22.2 KN) Tensile strength for snap hook and carabiners



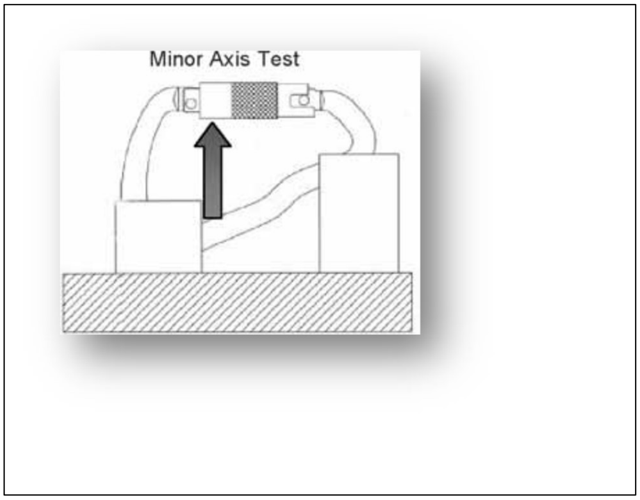

3,600 Lb. (16 KN) gate face strength rating




3,600 Lb. (16KN) side gate strength rating



3,600 Lb. (16KN) minor axis rating on non-captive eye snap hook and carabiner




All connectors **shall** be connected in such a manner where there is only one connection per connecting point unless transferring to another system. Care shall be taken to prevent the accidental disengagement of the connector through binding or rigged surfaces. Snap hooks shall meet OSHA and ANSI Z359.12(09) requirements.

## Anchor Straps and Slings

All anchorage connecting components intended to be used for fall protection **<u>shall</u>** be rated for fall protection and **<u>shall</u>** not be used for any other application. Anchorage connectors being used for fall arrest **<u>shall</u>** be rated for 5000lbs. It is important to use the anchorage connector within the scope of the manufacturer due to the forces that can be generated if the anchorage connector is attached improperly.

## Energy Absorbing Lanyards


An integral component to a fall arrest system, these lanyards have an integrated energy absorber that is designed to dissipate the energy generated during a fall through means of tearing or deforming over a predetermined length. The system shall limit the amount of force transmitted to the body to 1800lbs maximum arresting force and an energy absorber the deploys no more the 4ft. These energy absorbing lanyards shall meet ANSI Z359.13.

**Work Positioning Lanyard**

A lanyard with an adjustable rope grab, that is used to position allowing worker to free the hands to conduct work. The work positioning system shall not be considered a back-up (safety) and must be used in conjunction with a back-up device

## Self-Retracting Devices (SRD)


Self-retracting devices are a fall arrest device that mechanically locks when a fall occurs. These devices are advantages to use when clearance to the nearest hazard is limited. There are many different classes of these devices and employees **shall** verify the SRD being used is suited for the work to be performed.

Old Classifications (some still on system):

- Class A SRD (Fall limiter) Maximum arresting distance of 24in with an average arresting force of 1350lbs and max of 1800lbs.
- Class B SRD Maximum arrest distance of 54in with an average arresting force of 900lbs and max of 1800lbs.
- SRD-LE (Leading Edge) The integral energy absorber shall meet Z359.13 standard regarding personal energy absorbers.

New Classifications:

- Class 1 SRDs should only be used where the anchorage is located at a sufficient height above the walking/working surface such that the free fall distance should be two feet or less, and free fall and swing fall should not result in contact with a lower level or an object in the fall or swing path. The maximum arrest force shall not exceed 1,800 pounds (8kN), the average arrest force shall not exceed 1,350 pounds (6kN) and the arrest distance shall not exceed 42 inches (1,067mm) under ambient conditions.

- Class 2 SRDs may be used in applications where free fall may exceed six feet and contact with the structural edge is possible. These conditions should be eliminated, if feasible due to the vulnerability of the line material to cutting and abrasion in conditions where sharp or abrasive edges may be present. The maximum arrest force shall not exceed 1,800 pounds (8kN), the average arrest force shall not exceed 1,350 pounds (6kN) and the arrest distance shall be measured and included in labelling and in the manufacturer's instructions.

Note that SRD's have limitations on use and application and are generally designed for overhead use.

## Fall Arrestors


These devices are designed to mechanically lock onto a compatible rope when a fall occurs. They commonly come in either manual or automatic operation. A personal energy absorber can and is often connected to the arrestor allowing it to serve both fall arrest and fall restraint functions. (Note when being used for fall arrest a personal energy absorber must be used.)

- Automatic arrestors which travel along with the worker lock automatically when a fall occurs. These devices prevent "death grip" from occurring. As A best practice, automatic arrestors should be used when an employee is working vertically.
- Manual arrestors (rope grabs) are under spring tension causing the device to apply constant force on the rope unless manually released. These devices are best suited for horizontal work, where the device helps prevent unintentional adjustment. These devices do not prevent against "Death Grip".

- These devices **shall** meet the OSHA requirements and ANSI Z359.15

**Note:** "Death Grip" is the unintentional disengagement of a device that causes the locking function of the device to not engage during a fall.

## Horizontal Lifelines (HLL)

A HLL system can be an option to provide fall protection in difficult areas such as large tower bridges where distance between anchorage points are too far apart to reach or aren't readily available. While a HLL can have the advantage of making horizontal progression easier for users, the proper installation of the HLL, safe use, and the dynamics of the system need to be considered. In accordance with OSHA requirements and policy set forth by the Fall Protection Committee, HLLs used **shall** meet the following minimum guidelines: horizontal lifelines shall be designed, installed, and used, **under the supervision of a qualified person**, as part of a complete personal fall arrest system, which maintains a safety factor of at least two.

## Knots for Fall Protection and Rescue

Knowledge of proper knots is a critical part of fall protection and rescue. An improperly tied knot or a "granny" knot could lead to injury or fatality. Knot tying will be a part of regular training. The best way to remember how to tie knots is by repetition.

The two approved knots for supporting a human load during fall protection and rescue are the figure 8 and butterfly knot.

**Figure 8 knot:**

Figure 8 knot: A figure eight knot is a knot that is only used for pulling in one direction. It is tied so that the knot becomes a loop on the end of a rope.

**Butterfly Knot:**

Butterfly knot: A butterfly knot is a very versatile knot that can be used to put a loop anywhere along a rope. The loop created by the knot can be pulled in any direction. The knot can also be tied to bypass a damaged section in the rope and still maintain the strength of the rope.

# SECTION III

## FALL PROTECTION WORK PROCEDURES

## Fall Protection Planning Procedures

### Job Briefing

In accordance with APM rule J-1, Job Briefing, personal protective equipment including fall protection **shall** be covered in the job briefing. Prior to beginning work where fall protection is required the employee in charge of the job **shall** review with the crew what kind of fall protection will be used and how they will implement it.

### Rescue Plan

A means of rescue for a worker that is working aloft **shall** be readily available at the job site. There are many options for rescue, and which one will work best is a situation by situation call by the workers performing the rescue. Prior to the job, a rescue plan outline, based off the hazards associated with the job, location of the hazards and tools available to assist with the rescue **shall** be discussed during the documented Job Briefing.

## Fall Protection Work Procedures

### Ladders

Portable ladders are one of the most commonly used pieces of equipment in our daily tasks. They are handy, simple to use, very versatile, practical and effective. Portable ladders are used to perform a variety of jobs, yet more injuries are recorded every year from falling from ladders than any other elevated surface (roofs, scaffolds, balconies even stairs).

Although fall protection is not required for climbing or working from portable ladders (as well as fixed ladders less than 20ft in length), many incidents that have occurred during the use of portable ladders could have been prevented by following some basic principles as outlined below.

1. ***What condition is the ladder in?***

   Neglected ladders quickly become unsafe ladders: step bolts loosen, sockets and other joints work loose, and eventually the ladder becomes unstable. All ladders should be inspected for defects before each use. If the ladder is damaged, it must be removed from service until repaired or replaced. It must be immediately marked as defective or tagged with “Do Not Use” or similar language.

   - Clean and lightly lubricate moving parts such as spreader bars, hinges, locks, and pulleys
   - Inspect and replace damaged or worn components and labels according to the manufacturer’s instructions
   - Inspect the rails of fiberglass ladders for weathering, fiber bloom, and cracks
   - Keep the ladder away from heat sources and corrosive materials
   - Ensure that anti-slip footings are in place

### 2. *Which Ladder is the right one for the job?*

Most portable ladders are either non-self-supporting, such as an extension ladder, or self-supporting, such as a standard stepladder. There are also combination ladders that convert quickly from a stepladder to an extension ladder.

Extension ladders offer the greatest length in a general-purpose ladder. The ladder consists of two or more sections that travel in guides or brackets, allowing adjustable lengths. The sections must be assembled so that the sliding upper section is on top of the lower section. Each section must overlap its adjacent section a minimum distance, based on the ladder's overall length. The overall length is determined by the lengths of the individual sections, measured along the side rails. The table below shows the minimum overlap for two-section ladders up to 60 feet long.

| Ladder length | Overlap |
|---|---|
| Up to 36 feet | 3 feet |
| 36 to 48 feet | 4 feet |
| 48 to 60 feet | 5 feet |

Extension ladders **shall** only be used by one person at a time.

The standard stepladder has flat steps and a hinged back. It is self-supporting and nonadjustable. Standard stepladders should be used only on surfaces that have a firm, level footing such as floors, platforms, and slabs. They're available in aluminum, wood, or reinforced fiberglass and are intended to support only one worker at a time. **DO NOT stand on the top step**. Stepladders must have metal spreaders or locking arms and can't be longer than 20 feet, measured along the front edge of the side rails.

Using a standard stepladder in a closed position is not a safe practice because it's more likely to slip on surfaces such as concrete and wood than a straight ladder. Standard stepladders are designed to be used only when the spreader arms are open and locked. If a standard stepladder doesn't meet your needs, choose an appropriate straight ladder or a combination ladder.

### 3. *Am I using the ladder as the Manufacturer intended?*

The total length of an extension ladder should be seven to 10 feet longer than the vertical distance to the upper contact point on the structure — a wall or roofline, for example. **Never stand on the ladder rungs that extend above a roofline.**

Manufacturers give ladders duty ratings, based on the maximum weight they can safely support. The worker's weight plus the weight of any tools and materials that are carried onto the ladder must be less than the duty rating. Before you use a ladder, consider the maximum weight it will support.

## Scaffolding

For use of scaffolding, contact the Fall Protection Committee.

## Aerial Lift Equipment

Fall Protection shall be required when working from an aerial lift. Each aerial lift **shall** be outfitted with a means of self-rescue that **shall** be always carried in the bucket or basket.


### *Fall Arrest in an Aerial Lift*

A BPA approved full body harness and a 6-foot shock absorbing lanyard attached to both the dorsal D-ring and the aerial devices rated anchorage point. **CAUTION:** the use of fall arrest in an aerial device does not prevent the employee from being ejected from the bucket by a sling shot affect should the bucket hang up on an object and suddenly release.

**When planning a job, the use of aerial lifts will be given primary consideration for access to work heights.**

*When possible, the reduction of exposure to the hazard of falling **shall** be accomplished by using an aerial lift* as the first choice of accessing work heights.

When work is not able to be accessed using an aerial lift, climbing will be used as the secondary means of access to work heights.

### *Fall Restraint in an Aerial Lift*

A BPA approved full body harness, and a personal SRD attached to either the dorsal D- ring and the rated anchorage in the aerial device. This method reduces the chance of the worker from being ejected due to the slingshot effect should the bucket hang up on an object and suddenly release.

**NOTE**: When transferring to a structure using a personal SRD in an aerial lift, attach your work positioning device to the structure before climbing above the rim of the bucket. When transferring from the structure to the bucket remain attached to the structure with your work positioning device until your feet are firmly on the bucket floor, then attach SRD to your harness and disconnect the work positioning device from the structure. This is to ensure the clutching mechanism doesn't fail where the potential of a factor of two fall exists.

## Scissor Lifts

Fall protection is not required while operating a scissor lift unless the lift is equipment with a dedicated fall protection anchorage point.

## Vertical Movement

An approved fall arrest or work positioning system **shall** be always used while ascending or descending a structure. The task and number of trained workers ascending the tower will dictate the type of system to be used.

## Approved Vertical Movement Procedures

### *Y-lanyard connecting to step bolt flanges*

When ascending or descending a tower using the step bolt flanges you **shall** use a short (approximately 3') twin tailed lanyard with a shock absorber connected to the sternal D.

Benefits of using a short twin tailed lanyard connected to the sternal D

- Minimizes fall distances and fall forces
- Puts the worker in a position less likely to receive head trauma
- Puts the worker in a more defensive position to help reduce bodily injury
- Puts the worker in a final resting position that is more conducive for self-rescue

When using a short twin tail Y-style lanyard

- Connect only the center snap hook to the sternal D-ring fall arrest attachment point on the harness
- Do not attach the leg of the lanyard, which is not in use to the harness, except to attachment points specifically designated by the manufacturer for this purpose
- Do not modify the lanyard to create more than a 6 ft. (1.8m) free fall
- Do not allow the legs of the lanyard to pass under arms, between legs or around the neck
- Do not connect the snap hook back to itself in a choking fashion unless the manufacture states that it can be used in this manner

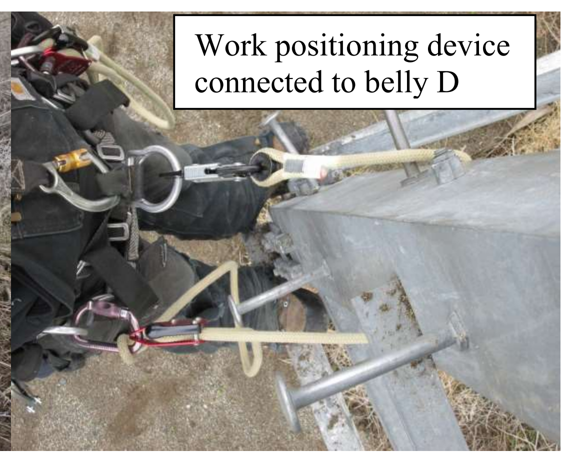
Y-lanyard connecting to step bolt flanges

## *Double Skidding*

Double skidding is defined as using two separate adjustable work positioning devices which are connected from one D-ring at the worker's hip to the opposing D-ring located at the other hip or the belly D-ring attachment. The trained worker shall remain 100 percent attached at all times.

Double skidding is considered work positioning, and rules apply accordingly. This method can be used where there are no step bolt flanges present on the tower, or where the number of climbers going up the structure makes the installation of a temporary lifeline impractical for the task being performed.

- When using the double skid method, the user needs to take ***special precautions as to which work positioning device is being attached and detached.***
- It is recommended that a visual identifier be used to distinguish the difference between the two work positioning devices if they are the same.
- The work positioning devices adjustable ends shall not attach to the same D-ring, this aids in the identification process between the two work positioning devices.
- Manage the slack rope on the work positioning device.
- Visually check the engagement of the snap hook into the D-ring.
- Visually check to ensure the snap hook being disconnected is the correct one.
- The use of the belly D-ring is also a way to make the connecting and disconnection of the snap hook end of the lanyard clearer and easier due to its orientation.

## *Temporary Vertical lifeline (VLL)*

The use of a temporary VLL requires a lead climber using a short Y style lanyard connected to a step bolt flange or using the double skid method to climb a structure while carrying and installing a lifeline to a rated anchorage point for the subsequent climbers.

- Use only BPA approved vertical life kit and components Catalog ID # **0001017517.**


- There **shall** be only one climber attached to the vertical lifeline at any one time.

- When calculating fall clearances be aware of rope stretch.

  Example: ½" ArmorTech @ 1,000 Lbf. = 8.7% elongation. This means if you have 100 feet of rope between the rope grab and the anchor, and a max arresting force (MAF) of 1,000 Lbs. the stretch of the rope will be approx. 8.7 ft.

- If no sewn eye is present on the rope, the figure eight knot with a carabiner shall be used for terminating the rope to an anchor, this is because the figure eight will not come untied easily and it retains 75% of the rope strength.

- Always inspect the components of the system before installation.

- When installing a vertical lifeline, the first person up **shall** always maintain attachment. The VLL can be redirected to follow the climbing access route as the first climber up gets to those locations or it can be redirected by the second climber once it is secured to an anchorage point.

  NOTE: when redirecting the lifeline as the second climber is ascending the structure there is a possibility of excessive swing fall due to the change in climbing angle. It also may be necessary to redirect the lifeline as the first climber ascends the tower to maintain MAD distances.

First person up redirecting as he climbs.

- Always check the orientation, operation, and rope size requirements of the rope grab assembly before ascending or descending on the system. This can be done by placing the rope grab on the system then quickly sliding the grab in the direction of the anticipated fall; the grab should lock up within the distance specified by the manufacture. After the test is performed make sure the grab is reset and operates smoothly in both directions.

- During the installation and retrieval of the VLL, special precautions should be taken to ensure clearance from energized conductors is maintained.

- The vertical lifeline should be installed and redirected to follow the climbing angle but be off the step bolts to avoid tripping hazards.


### *Structures with permanently installed rung ladders as the primary means of access*

For structures with permanently installed rung ladders as the primary means of access, a trained worker **shall** attach a pelican hook around a rung and the rail as the approved anchorage. The trained worker **shall** use a pelican hook in conjunction with the short Y style lanyard or a work positioning device while ascending the structure.

Do not use any permanent system, such as vertical lifelines, ladder rail systems, or any other system that has not had regular inspection intervals prior to 4/1/2015.

### Horizontal Movement

When moving horizontally across a structure, trained workers **shall** be attached at all times.

The trained worker shall select a method that minimizes fall distances where a lower level and energized conductors are a recognized hazard.

### Approved Horizontal Movement Procedures

#### Double Skidding:

Double skidding is defined as using two separate adjustable work positioning devices which are connected from one D-ring at the worker's hip to the opposing D-ring located at the other hip or the belly D-ring attachment. The trained worker shall remain 100 percent attached at all times.

Double skidding is considered work positioning, and rules apply accordingly. This method can be used where there are no step bolt flanges present on the tower, or where the number of climbers going up the structure makes the instillation of a temporary lifeline impractical for the task being performed.

- When using the double skid method, the user needs to take ***special precaution as to which work positioning device is being attached and detached.***

- It is recommended that a visual identifier be used to distinguish the difference between the two work positioning devices if they are the same.

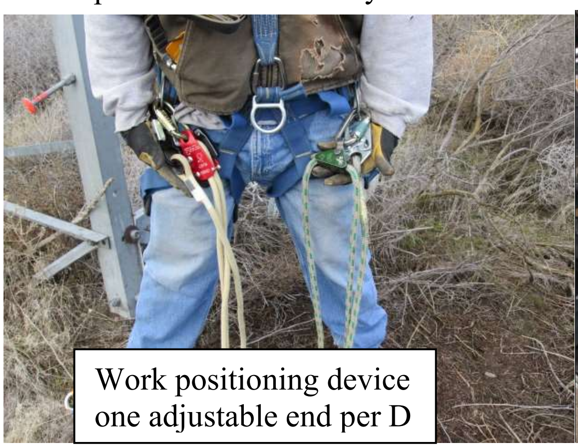
- The work positioning devices adjustable ends shall not attach to the same D- ring, this aids in the identification process between the two work positioning devices.

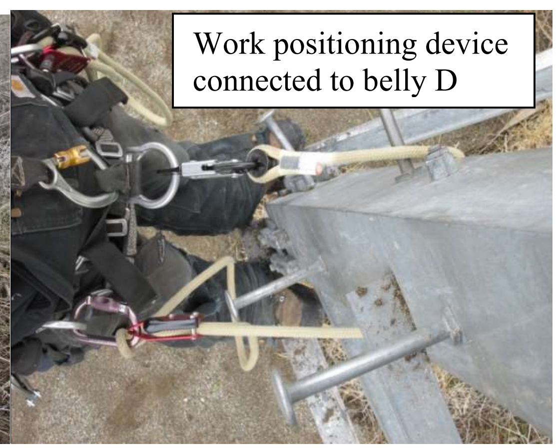
- Manage the slack rope on the work positioning device.

- Visually check the engagement of the snap hook into the D- ring.

- Visually check to ensure the snap hook being disconnected is the correct one.

- The use of the belly D-ring is also a way to make the connection and disconnection of the snap hook end of the lanyard clearer and easier due to its orientation.



**Pelican Hooks Connected to a Short Y Lanyard**
This method is considered a fall arrest system, and rules apply accordingly.

When using a short twin tail Y-style lanyard ensure:

- Connect only the center carabiner to the sternal D-ring fall arrest attachment point on the harness.
- Do not attach the leg of the lanyard, which is not in use to the harness, except to attachment points specifically designated by the manufacturer for this purpose.
- Do not modify the lanyard to create more than a 2 ft.free fall.
- Do not allow the legs of the lanyard to pass under arms, between legs or around the neck.
- Do not connect the snap hook back to itself in a choking fashion unless the manufacturer states that it can be used in this manner.
- Special precautions should be taken to calculate fall clearances to avoid hitting a lower level.
- Always connect pelican hooks to steel members or anchorages that meet fall arrest requirements (see anchorages section).
- Be aware of connecting to steel members which will cause the pelican hooks to slide to the workers feet resulting in more than a six-foot free fall.

## Pelican Hooks Connected to an Adjustable Work Positioning Device

This method is considered fall restraint and rules apply accordingly.

- Allows trained workers to stay connected when moving horizontally where an overhead anchorage is available and the span lengths between anchorages are too long to use the double skid method.
- The adjustable work positioning device shall be connected to the belly or sternal D-rings only.
- Keep work positioning device adjusted as short as possible.
- Adjustable work positioning devices must be rated by the manufacturer to support human loads in the vertical configuration (RAD lanyard).
- The anchorage under fall restraint must be able to withstand 1000 Lbs.

## Horizontal Lifelines (HLL)

A HLL system can be an option to provide fall protection in difficult areas such as large tower bridges where distance between anchorage points is too far apart to reach or aren't readily available. While a HLL can have the advantage of making horizontal progression easier for users, the proper installation of the HLL, the safe use of it, and the dynamics of the system need to be considered. In accordance with OSHA requirements and policy set forth by the Fall Protection Committee, HLLs used **shall** meet the following minimum guidelines:

- Prior to using or installing a HLL, workers **shall** be trained.
- HLLs **shall** be installed and used under the supervision of a qualified person.
- All components of the HLL system **shall** be inspected for integrity prior to use.
- Prior to installation of the HLL, trained workers **shall** consider fall clearances and hazards associated with the use of the HLL at that worksite.
- HLLs **shall** be anchored to locations on structures that have been identified by a qualified person as acceptable for use as HLL anchorages. Approved anchor straps **shall** be used to connect the system to the proper anchorage points.
- Correct installation of the HLL system **shall** be verified by a qualified person prior to relying on the system for fall protection.
- As with all work at heights, a rescue plan involving use of a HLL system **shall** be in place prior to the start of work.

## Wood Poles

Use of fall protection on wood poles is always mandatory when a trained worker (WPFRD) is working at heights above four feet. A WPFRD **shall** be used in conjunction with a full-body harness while climbing or working from a wood pole unless an alternate method of fall protection has been specifically approved by the Fall Protection Committee. To facilitate rescue, all trained workers **shall** wear BPA approved Fall Protection PPE while climbing or working.

The WPFRD **shall** be placed around the pole before the second step is taken and **shall not** be removed until back on the ground, except while passing across or around obstructions while maintaining 100% attachment using a secondary device placed above the obstacle to maintain fall protection. The WPFRD **shall not** be placed around a pole above the top cross arm except where adequate precaution is taken to prevent the strap from slipping over the top of the pole.

Prior to climbing a wood pole, an inspection **shall** be made by the trained worker for shell rot or other defects to determine that the structure can sustain the additional or unbalanced loads to which they will be subjected. Where poles or structures may be unsafe for climbing, they **shall** be maintained by use of an aerial devise or **shall not** be climbed until made safe by guying, bracing, or other adequate means of support.

Sliding X-braces or guy wires **shall** be prohibited.

The use of hand axes or power saws on overhead work where employees are supported by a WPFRD **shall** be prohibited unless wire rope is used in conjunction with the WPFRD.

## Transferring To and From Elevated Work Locations

A trained worker may transfer between any combination of the following: Aerial Lift, Structure, Conductor, Aerial Ladder, Cable Cart, or Electrical Equipment in accordance with the following:

- Buckets and platforms **shall** be designed to remain stable during transfer. The bucket or platform **shall** have a fixed pin or a locking device when required to provide stability during transfer.

- Transfer from an aerial lift **shall** be made from the device by a door, or step designed solely for the purpose of assisting the trained worker over the rim of the bucket.

- Protective covers **shall** be in place on pole climber gaffs while working from an aerial lift, aerial ladder, or aerial cart.

- A climber transferring between an aerial lift and a structure, or suspended platform and structure, **shall** be attached to either the aerial device or structure while making the transfer, minimizing the time connected to both.

- During the transfer of the attachment, the trained workers feet **shall** be firmly planted on the floor of the aerial device or on the structure before disconnecting.

- When transferring using a personal SRD in an aerial lift, attach your work positioning device to the structure before climbing above the rim of the bucket. When transferring from the structure to the bucket remain attached to the structure with your work positioning device until your feet are firmly on the bucket floor then attach SRD to your harness and disconnect work positioning device from the structure. This is to ensure the clutching mechanism doesn't fail where the potential of a factor of 2 fall exists.

### Suspended Work Locations

Trained workers **shall** be attached using fall restraint or fall arrest at all times when working from suspended platforms, ladders, boatswain's chair, carts etc. During the transfer between the structure or lift equipment and the suspended work location, the trained worker shall stay attached at their location until they are attached at their new location minimizing the time attached to both.

When positioning for work in suspended work locations, consider the best location/body position for the work at hand. Ergonomics and proper body positioning are key in reducing/eliminating ergonomic injuries while working at height.

### Hook Ladders

The use of vertical ladder lifelines or SRD are the approved hook ladder fall protection procedures.

Only use BPA approved vertical ladder lifeline kit and components Catalog ID # **0001015733.**

A trained worker moving from a structure to a hook ladder **shall** ensure the ladder is properly positioned and secured to the structure by means of safety chains. The trained worker will then install their vertical lifeline (VLL) or self-retracting device (SRD) to a proper anchorage. The trained worker **shall** remain attached to the structure until they have connected their harness to either the VLL or SRD via the dorsal or sternal D-ring. Once they are connected, they may remove their means of attachment to the structure and climb onto the ladder.

Once on the ladder, the qualified employee may descend to their desired work location. Once at their work location, the worker will place their work positioning device around the ladder.

At no time during the course of work, climbing up or down the ladder, **shall** the VLL or SRD be detached from the qualified employee. To transfer from a ladder to a structure, this procedure is reversed.

### Spacer Carts

While in a spacer cart, a work positioning pole strap attached to the side D-rings of the user's harness **shall** be used around an anchorage such as a conductor, insulator, yoke, etc., and is the only approved method of fall protection. Two work positioning devices **shall** be used to maintain attachment while transitioning around or over obstacles while in the spacer cart.

### Dead End boards

While working on dead-end boards the employee **shall** stay always connected. Due to the location of this work, it is difficult to maintain attachment to an approved anchorage. For this reason, it is permissible to use a work positioning device around rigging that is used to tension insulators. The use of two work positioning devices, one always connected either around insulators or rigging, has been determined to be the best method in this situation.

The use of rope installed around the edge and through the end of the dead-end board for fall protection is prohibited. The dead-end board and rope are not of sufficient strength to use as an anchor and cause a greater risk of additional employees falling.

## When Fall Protection is Infeasible

OSHA's new ruling on fall protection 1910.269(g)(2)(iv)(C)(3) There is the following verbiage:

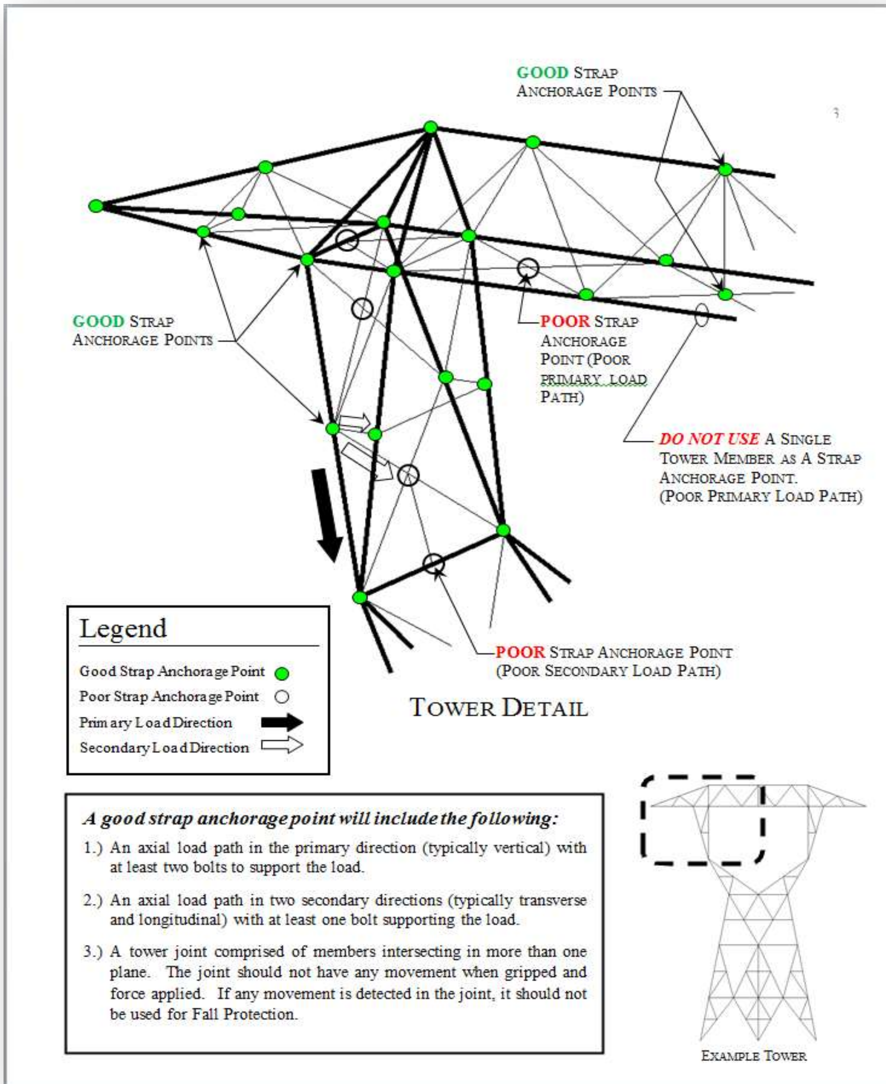
> *On and after April 1, 2015, each qualified employee climbing or changing location on poles, towers, or similar structures must use fall protection equipment unless the employer can demonstrate that climbing or changing location with fall protection is infeasible or creates a greater hazard than climbing or changing location without it.*

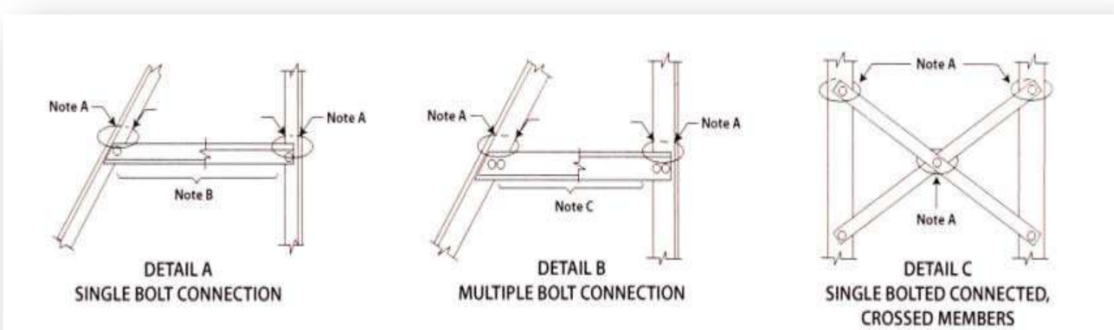
It is recognized that there may be situations based off of structure design limitations or work procedures to be performed where the use of currently available fall protection gear or procedure may create a greater hazard than the hazard it is designed to protect from. This verbiage in OSHA is designed solely for safety reasons. Inconvenience, additional time to complete the work procedure or failure to plan for specific fall protection needs to complete a work procedure does not meet the" infeasible definition". The use of "infeasible" will be evaluated on a situation-by-situation basis as they arise by the fall protection committee and safety office.

## Anchorages

### Anchorages – Legacy Towers

For legacy towers, a guide has been created by BPA Structural Design Group that identifies anchorage points approved for a maximum 3000 Lb. vertical fall protection load. These are calculated anchorage points capable of withstanding twice the maximum arresting force anticipated in a single fall arrest system (<1500Lbs) with a safety factor of 2:1 as defined in OSHA 1026.502(d)(15). These anchorage points are typically places where three or more tower members intersect. The following text and drawings provide a guideline to follow when identifying these anchorage points on legacy towers.

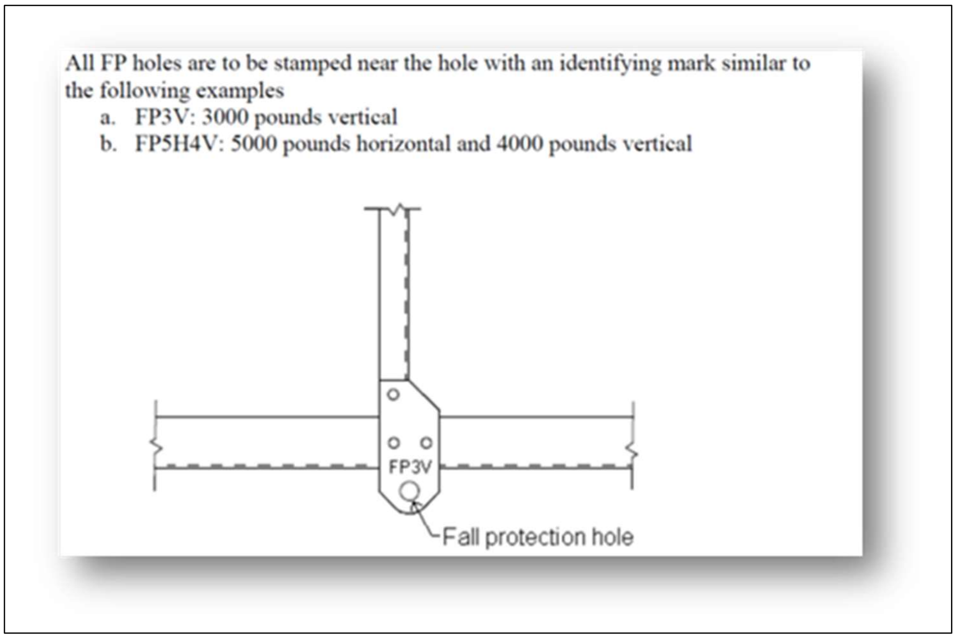

GOOD STRAP ANCHORAGE POINTS
GOOD STRAP ANCHORAGE POINTS
POOR STRAP ANCHORAGE POINT (POOR PRIMARY LOAD PATH)
DO NOT USE A SINGLE TOWER MEMBER AS A STRAP ANCHORAGE POINT. (POOR PRIMARY LOAD PATH)
POOR STRAP ANCHORAGE POINT (POOR SECONDARY LOAD PATH)
TOWER DETAIL
Legend
Good Strap Anchorage Point
Poor Strap Anchorage Point
Primary Load Direction
Secondary Load Direction
A good strap anchorage point will include the following:
1.) An axial load path in the primary direction (typically vertical) with at least two bolts to support the load.
2.) An axial load path in two secondary directions (typically transverse and longitudinal) with at least one bolt supporting the load.
3.) A tower joint comprised of members intersecting in more than one plane. The joint should not have any movement when gripped and force applied. If any movement is detected in the joint, it should not be used for Fall Protection.
EXAMPLE TOWER

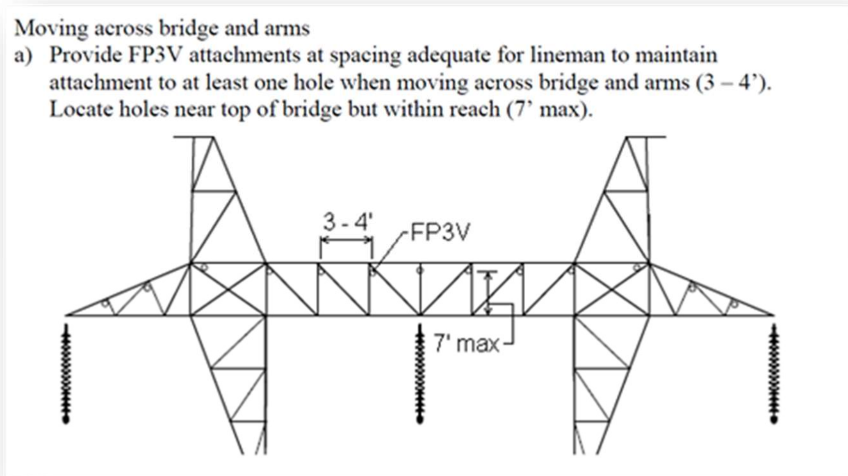

Note A
Note A
Note B
DETAIL A
SINGLE BOLT CONNECTION
Note A
Note A
Note C
DETAIL B
MULTIPLE BOLT CONNECTION
Note A
Note A
DETAIL C
SINGLE BOLTED CONNECTED, CROSSED MEMBERS

Notes:
A. Attachment around and above all joints is permitted.
B. Attachment around a lattice member supported by one bolt on each end may be permitted with the proper work positioning equipment.
C. Attachment around multiple bolted members is permitted.

**Anchorages - New Towers**

Towers designed post-2013 will have engineered anchorage points on the structure, improved anchorage location and step bolts with flanges. See examples below.






**Tower – FP Anchorage Point Drawings**
http://internal.bpa.gov/Policy/Safety/Pages/Safety.aspx
See the Resource section of Fall Protection Handbook.

## Power System Equipment Fall Protection Work Procedures

### Crane/Derrick

A Crane/Derrick may be used as a fall protection anchorage point provided the load chart is referenced for a minimum of 3600 Lb. load.

Anchoring to the load line - A personal fall arrest system is permitted to be anchored to the crane/derrick's hook or other part of the load line where the following requirements are met:

- A competent person (crane and fall protection) has determined that the set-up and rated capacity of the crane/derrick (including the hook, load line and rigging) meets or exceeds 3600 lb.
- No load is suspended from the load line when the personal fall arrest system is anchored to the crane/derrick's hook (or other part of the load line).
- Lockout/Tagout procedures on the craned/derrick employed

## Miscellaneous Anchor Points

Finding acceptable anchorage points is often a challenge for fall protection. There is no way to list all available materials and systems in this manual. There are many commercial systems available on the open market. Special needs anchorage systems **shall** be approved by the Fall Protection Committee.

The following are some examples of available anchorage systems:

## Fall Arrest Post

This product has multiple mounting options for its base unit. Base units that bolt on, clamp on or require welding are all available. The post is attached to the base for use as an anchorage point. The post can be removed and transferred to other properly attached bases. For substations the District Foreman **shall** ensure that all transformers and reactors within the district have the plate for the fall arrest post welded onto the transformer. The plate **shall** be welded onto the transformer by a BPA certified welder.

## Free Standing Anchorage

Free standing anchorage systems are available from multiple manufacturers. These systems are a weight-based system that can be lifted onto a rooftop or similar platform and provide a temporary anchorage.

# SECTION IV
## MISCELLANEOUS

**Resources**

**Anchorage Drawings for Legacy Towers**
As Engineering completes their review of legacy tower designs for anchorage point, the drawings will be posted to the Fall Protection section of the Safety SharePoint site.

**SharePoint**
Fall protection information is located on the Safety SharePoint site.
http://internal.bpa.gov/Policy/Safety/Pages/Safety.aspx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